



SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

DEPARTMENT OF MECHANICAL ENGINEERING, SRMIST, KTR

In association with ISHRAE

Organises

SRM Smart Park Hackathon

EXHIBITION OF THE WORKING MODELS

Date: October 28th, 2025

Venue: Mechanical Hangar, SRMIST, Main Campus, Kattankulathur

Design a multi-level smart car parking system that integrates edge-based IoT sensing, computer vision for vehicle detection, and cloud-driven predictive analytics to optimize parking space usage in real time. The system must dynamically allocate slots based on vehicle type, and priority rules while enabling secure user authentication and payment integration.



Technical Constraints:

- ★ **Slot Capacity & Sensing:** System must simulate a minimum of 100 parking slots using IR/ultrasonic sensors or computer vision for real-time detection with $\geq 90\%$ accuracy.
- ★ **Cloud & Analytics:** Sensor data must be logged to a free-tier cloud platform and include at least one ML model to forecast slot availability or parking demand.
- ★ **Interface & Access:** A mobile/web app must support real-time slot view, booking/cancellation, and secure user authentication.
- ★ **Power & Performance:** Edge devices should support low-power (battery, solar) operation with ≤ 5 sec data latency and offline data caching capability.

Scalable up to 100 parking slots with $\geq 90\%$ accuracy, use at least one ML model to forecast slot availability, must support real-time slot view, booking/cancellation, and secure user authentication on a mobile/web app, must be battery powered.

A team of up to 5 students, from various disciplines of Engineering & Technology are required to participate.

₹400

1st

Prize

₹ 8000/-

2nd

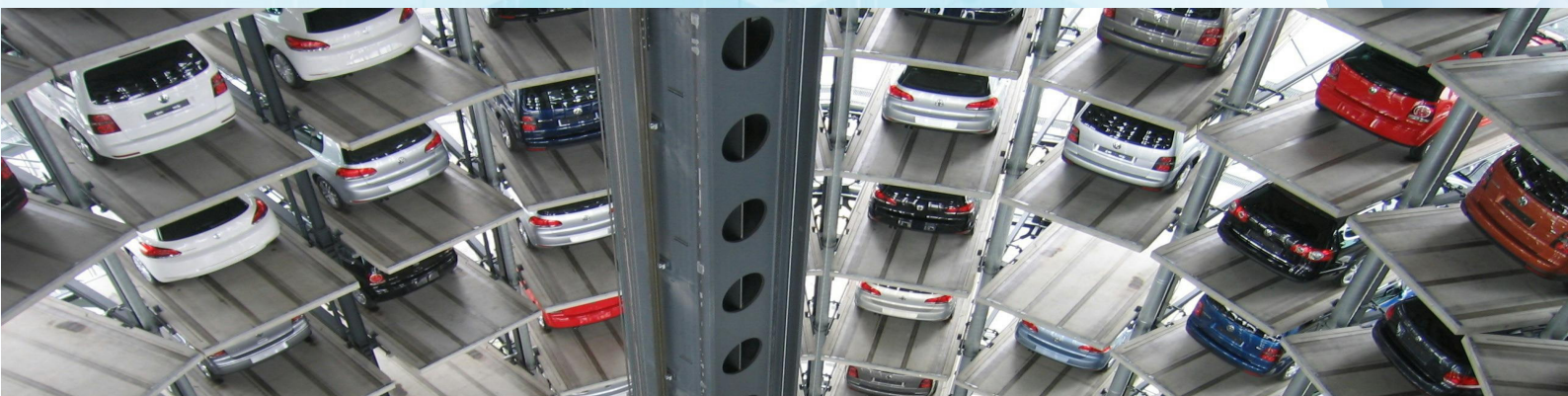
Prize

₹ 5000/-

3rd

Prize

₹ 3000/-



Last Date for Registration :	23.10.2025
Team Confirmation & Rubrics :	24.10.2025
Date of Event (Final Demo) :	28.10.2025

Dr. K. Suresh Kumar,
Professor & Head
Dr. V. Praveena,
Associate Professor

Dr. P. Chandrasekaran,
Professor

Dr. Deborah Serenade Stephen,
Assistant Professor

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Account No.: **117001000635544**
Account name: **ISHRAE STUDENT CHAPTER - SRM UNIV**
Bank Name: **City Union Bank**
Account type: **Savings Bank A/C.**
IFSC code: **CIUB0000117.**



<https://tinyurl.com/srm-park>